EXPERIMENT 7

Aim: To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory:

Ganache is a personal Ethereum blockchain used for testing and development. It allows developers to deploy contracts, develop decentralized applications (DApps), and run tests in a safe and deterministic environment.

Ganache provides:

- Instant mining of blocks
- Pre-funded test accounts
- Transaction history view
- Graphical interface to monitor blockchain activities

It is part of the Truffle Suite and is commonly used with Remix and Metamask for smart contract deployment.

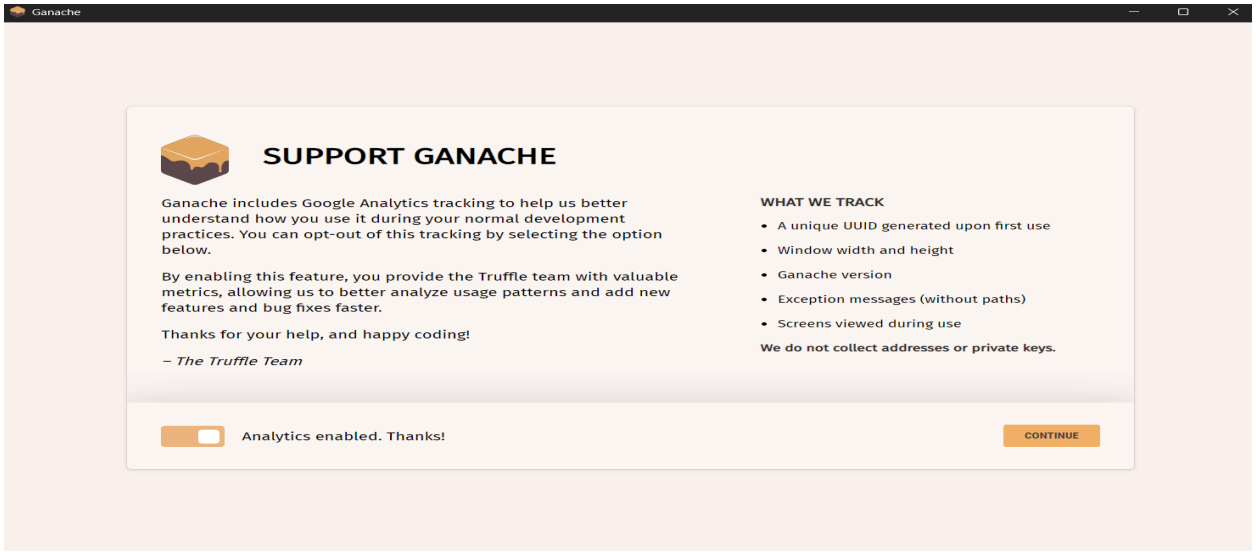
Steps to Connecat Ganache with Metamask and Remix IDE:

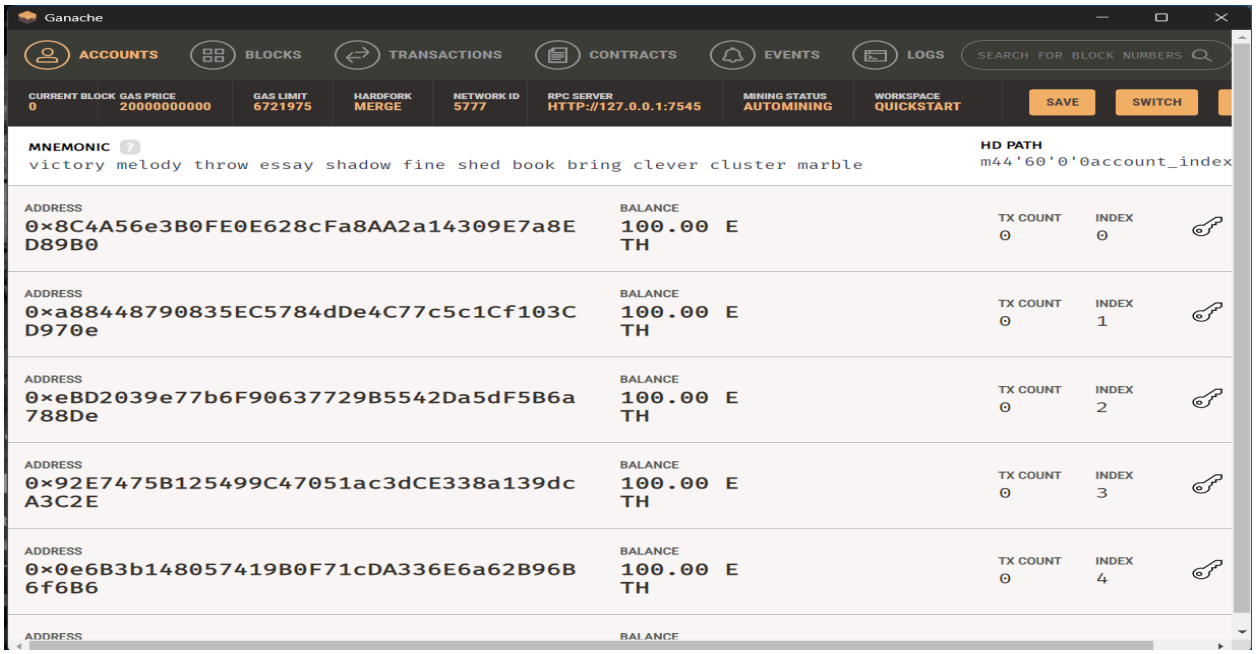
1. Install Ganache:
    - Download and install Ganache from the Truffle Suite website.
    - Launch it to start a local blockchain network with test Ether accounts.
2. Connect Ganache Accounts with Metamask:
    - Open Metamask and switch to a Custom RPC Network.
    - Enter the Ganache RPC server URL (e.g., HTTP://127.0.0.1:7545).
    - Import private keys of Ganache accounts into Metamask.
3. Connect Remix IDE with Metamask:
    - Open Remix IDE in the browser.
      Choose Injected Web3 as the environment in the “Deploy & Run Transactions” tab.
    - Remix will automatically connect to the Metamask account (linked with Ganache).
4. Create a Simple Solidity Smart Contract:
    - Write a Solidity program related to your mini project (e.g., Student Record, Donation Tracker, etc.).

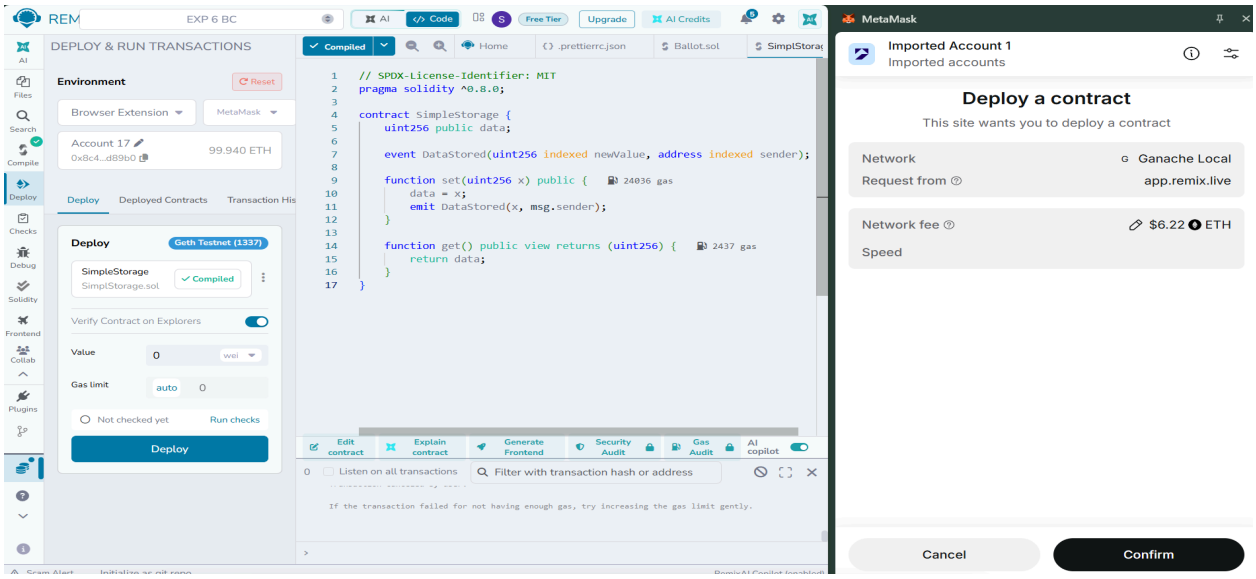
```
pragma solidity ^0.8.0;
    contract SimpleStorage {
      uint public data;
      function set(uint x) public {
        data = x;
      }
```

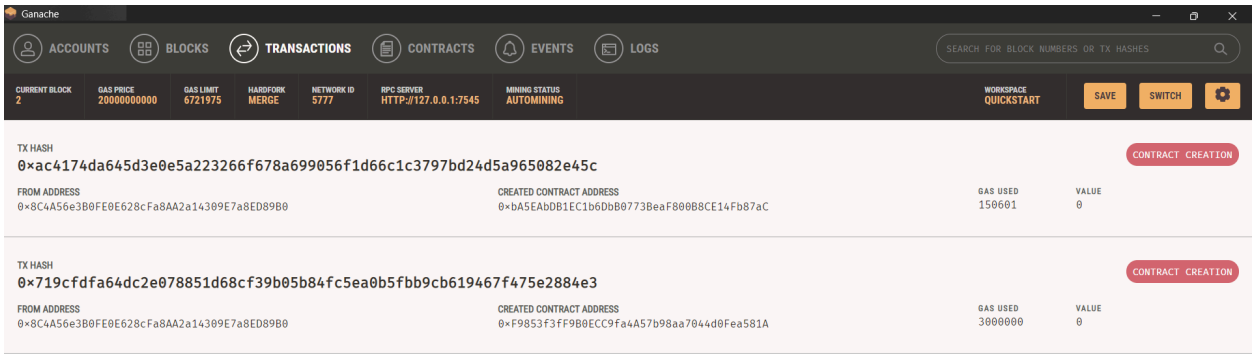
5. Compile and Deploy the Contract:
   Compile using Remix.
   - Deploy via the Metamask account connected to Ganache.
   - Approve the transaction in Metamask.
6. Check Transaction Details on Ganache:
   - View mined transactions, block details, and balances in the Ganache interface.
7. Interact with the Smart Contract:
   - Use Remix to call contract functions (set() or get()).
   - Observe the state changes and corresponding transactions on Ganache.

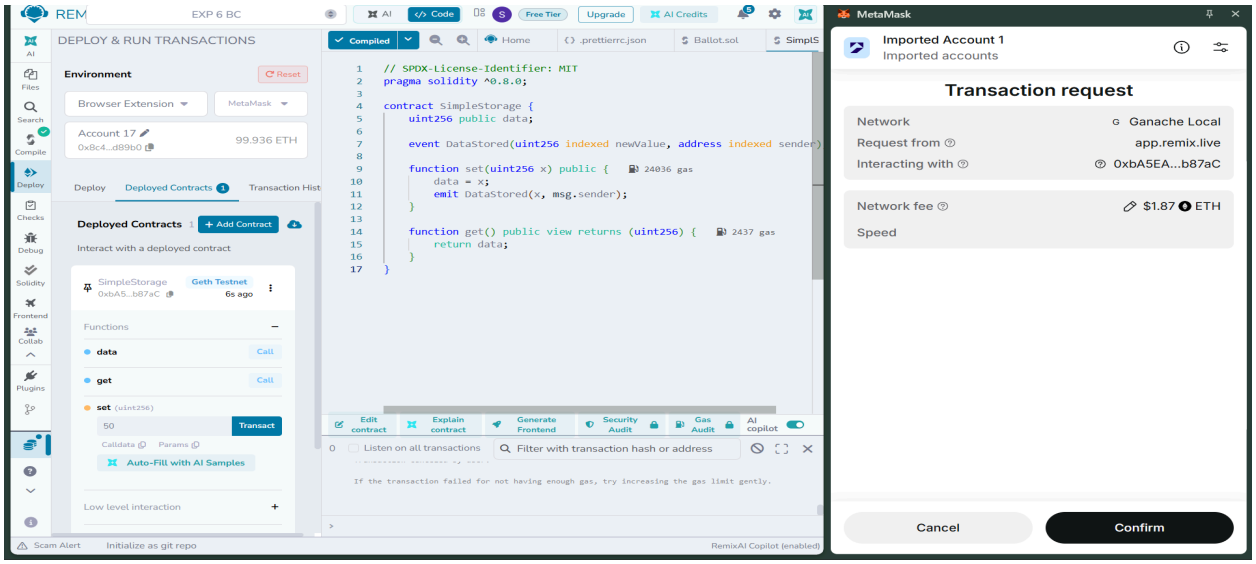
Output & Code:






DEPLOY & RUN TRANSACTIONS
Environment
Browser Extension
Account 17
0x8c4...d89b0
99.940 ETH
Deploy
Deployed Contracts
Transaction His
Deploy
Geth Testnet (1337)
SimpleStorage
SimplStorage.sol
Compiled
Verify Contract on Explorers
Value
0
wei
Gas limit
auto
0
Not checked yet
Run checks
Deploy
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract SimpleStorage {
uint256 public data;
event DataStored(uint256 indexed newValue, address indexed sender);
function set(uint256 x) public {
24036 gas
data = x;
emit DataStored(x, msg.sender);
}
function get() public view returns (uint256) {
2437 gas
return data;
}
}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
If the transaction failed for not having enough gas, try increasing the gas limit gently.
MetaMask
Imported Account 1
Imported accounts
Deploy a contract
This site wants you to deploy a contract
Network
Ganache Local
Request from
app.remix.live
Network fee
$6.22 ETH
Speed
Cancel
Confirm


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
CURRENT BLOCK 2
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART
SAVE
SWITCH
TX HASH
0xac4174da645d3e0e5a223266f678a699056f1d66c1c3797bd24d5a965082e45c
CONTRACT CREATION
FROM ADDRESS
0x8C4A56e3B0FE0E628cFa8AA2a14309E7a8ED89B0
CREATED CONTRACT ADDRESS
0xbA5EAbDB1EC1b6DbB0773BeaF800B8CE14Fb87aC
GAS USED
150601
VALUE
0
TX HASH
0x719cfdfa64dc2e078851d68cf39b05b84fc5ea0b5fbb9cb619467f475e2884e3
CONTRACT CREATION
FROM ADDRESS
0x8C4A56e3B0FE0E628cFa8AA2a14309E7a8ED89B0
CREATED CONTRACT ADDRESS
0xF9853f3fF9B0ECC9fa4A57b98aa7044d0Fea581A
GAS USED
3000000
VALUE
0


DEPLOY & RUN TRANSACTIONS
Environment
Browser Extension
Account 17
0x8c4...d89b0
99.936 ETH
Deploy
Deployed Contracts 1
Transaction Hist
Deployed Contracts 1
+ Add Contract
Interact with a deployed contract
SimpleStorage
0xbA5...b87aC
Geth Testnet
6s ago
Functions
data
Call
get
Call
set (uint256)
50
Transact
Calldata
Params
Auto-Fill with AI Samples
Low level interaction
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;
contract SimpleStorage {
uint256 public data;
event DataStored(uint256 indexed newValue, address indexed sender);
function set(uint256 x) public {
24036 gas
data = x;
emit DataStored(x, msg.sender);
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return data;
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}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
If the transaction failed for not having enough gas, try increasing the gas limit gently.
Scam Alert
Initialize as git repo
MetaMask
Imported Account 1
Imported accounts
Transaction request
Network
Ganache Local
Request from
app.remix.live
Interacting with
0xbA5EA...b87aC
Network fee
$1.87 ETH
Speed
Cancel
Confirm

Conclusion:

In this experiment, a local Ethereum blockchain was successfully set up using Ganache and integrated with MetaMask and Remix IDE. A Solidity smart contract (`SimpleStorage`) was compiled, deployed, and executed on the local network. Inspecting the mined blocks, transaction hashes, and updated account balances in Ganache demonstrated how smart contract deployments and state-changing transactions operate on an Ethereum blockchain in a controlled local testing environment.